

OD3 Neuro Practical Review Session

Agenda

30 min Expectations (Likert Scale) + Q&A

1 hr Practice case that you did not go over before

30 min Practice patient presentations with one another

Timing

1 min	Review doorway sheet outside room Includes vital signs, PMHx, and 6-point treatment plan outline. Make your history taking template.
17 min 9 & 2 min remaining notifications	History & Physical Faculty will provide pertinent “H&P” cards. Talk directly to the patient and pretend faculty is not there. Do not break role to ask for cards.
	HARD STOP
10 min 6 and 2 min remaining notification	Collect thoughts AND present assessment and plan to the Faculty Recommend starting your presentation shortly after the 6 minute and no later than the 2 minute remaining overhead announcements. You will only be graded for what you verbally present.
	HARD STOP
4 min	Faculty feedback
1 min	Exit

History

Summarize history and confirm accuracy with patient before moving onto physical

Empathy: To save time, take one empathy statement followed by a history question so the patient doesn't spend a lot of time responding to that statement.
EX: "I'm so sorry to hear [that you have a headache]. How often does this occur?"

To gain full marks on organization, ask all HPI ("BOLDCARTS") first followed by associated systems, then PMHx, Meds/Allergies, SHx etc

Physical Exam Tips

Start with your basic NMSK screening exam from 1st year. Based on what is abnormal and your history, choose your NMSK special tests accordingly.

Make sure you assess the following in your initial NMSK screening PE

- Mental Status
- Cranial Nerves
- Motor/Motion
- Coordination
- Reflexes
- Sensory
- Gait (If regular gait is normal, no need to do heel walking, toe walking, etc)

Physical Exam Tips

Fundoscopic exam:

- Make sure the light is on the pupil
- Make sure you get close enough
- If you have a hard time seeing the retina, you do not have to spend excessive amounts of time to do so. Similarly, if you are unable to find a reflex, you may move on after a couple attempts assuming you appropriately demonstrated how to set it up

Coach patient to take quick breaths for lung auscultation as if they're blowing out a candle

The presentation

To faculty

Use medical jargon

Assessment and plan only

1. Summary sentence
2. Working diagnosis + 4 supporting reasons (2 from HPI, 2 from PE)
3. 3 Differentials + 1 reason why you considered and 1 reason why less likely
4. 6 point plan for problem 1 (ie. working dx)
5. Plan (doesn't have to be 6 points) for somatic dysfunction

Differentials

1. Post Concussion Syndrome
2. Subarachnoid Hemorrhage
3. Meningitis
4. Brain Tumor
5. Pseudotumor Cerebri (Idiopathic Intracranial Hypertension)
6. Temporal Arteritis
7. Cluster Headache
8. Migraine Headache
9. Occipital Neuralgia
10. Tension Headache
11. Medication overuse headache
12. Trigeminal Neuralgia
13. Whiplash Injury Headache (cervical strain or cervicogenic headache)
14. Normal Pressure Hydrocephalus
15. Ischemic Stroke
16. Spinal Cord Compression

Examples of different physical exams
depending on the differential

Post-concussion syndrome

Goal of physical exam: look for red flags ie. focal neurological deficits

- Mental status exam (alertness and orientation)
- Cranial nerves
 - Visual field confrontation - to make sure they can see
 - Extraocular eye movements
 - Fundoscopic exam - looking for papilledema ie. increased intracranial pressure
 - CN VII - smile, close your eyes, raise forehead, look for facial asymmetry
- Tone
- Strength
- Sensation
- Gait (normal and tandem walking ok)
- Reflexes

Normal pressure hydrocephalus

Goal of physical exam: Do they fit the classic triad of wet, wacky, wobbly? Can we rule out Parkinson's and spinal cord compression?

- Mental status exam (alertness and orientation)
- Cranial nerves II-XII
 - Including fundoscopic exam to rule out increased intracranial pressure
- Tone (to rule out Parkinson's and assess for UMN or LMN signs)
- Strength (especially in lower extremities to rule out spinal cord compression)
- Sensation (LMN sign, especially in lower extremities to rule out spinal cord compression)
- Gait (normal and tandem walking)
- Reflexes (to look for UMN or LMN signs)
- Special tests to rule out UMN lesions, especially in the LE to rule out spinal cord compression
 - Babinski

Practice time

17 min Student 1 performs H&P

4 min for student 1 to gather thoughts

1 min switch

17 min Student 2 performs H&P

4 min for student 2 to gather thoughts

10 min for everyone to fill out clinical reasoning sheet.

Reconvene as a group and discuss cases.

Per likert scale

Basic examination technique skills (e.g. correct bracing, positioned on the right, listening posts, ladder fashion, auscultate entire cardiac and resp cycle, correct order of auscultation and palpation/percussion , etc.)

Strength: at least shoulder abduction, elbow flexion/extension, hip flexion, knee flexion/extension, ankle flexion/extension

Sensation: Face V1, V2, V3

Proximal and Distal in each extremity. Only light touch necessary.

**Special Tests –
Indications for
performance**

Fundoscopic exam

**Visual Fields (four
quadrants)**

**Extra-ocular
movements**

**Facial strength
(open/close eyes
tightly, smile)**

Pronator Drift

Finger-nose-finger

**Reflexes (biceps,
triceps,
brachioradialis,
patella, achilles)**

Gait

Kernig/Brudzinski